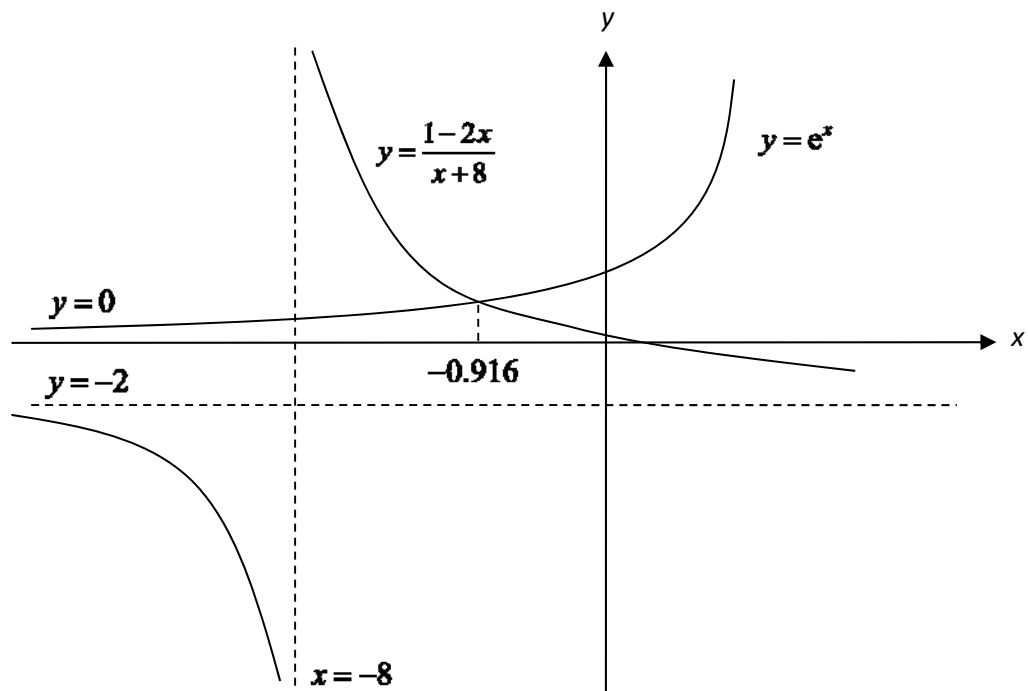


Solution

1(i)



(ii) $x < -8$ or $x > -0.916$

$$\frac{e^x(1+2x)}{8-x} < 1$$

$$\frac{1+2x}{8-x} < e^{-x} \quad (\text{since } e^x > 0)$$

Replace x by $-x$,

$$-x < -8 \quad \text{or} \quad -x > -0.916$$

$$x > 8 \quad \text{or} \quad x < 0.916$$

2(i) $\frac{dy}{dx} = 3x^2 + 8x + k$

At $x = 1$, $\frac{dy}{dx} = 11 + k$ and $y = 5 + k + k^2$

Equation of tangent at P :

$$y - (5 + k + k^2) = (11 + k)(x - 1)$$

$$y = 11x - 11 + kx - k + 5 + k + k^2$$

$$y = (11 + k)x + k^2 - 6$$

(ii) Sub $(2, 1)$ into the equation of tangent,

$$1 = (11 + k)(2) + k^2 - 6$$

$$k^2 + 2k + 15 = 0$$

Using discriminant, $b^2 - 4ac = 2^2 - 4(15)$
 $= -56 < 0$

Since discriminant < 0 , there are no values of k such that the tangent passes through the point $(2, 1)$.

3(i) $2xy + \pi x^2 = 2500\pi$

$$2xy = 2500\pi - \pi x^2$$

$$y = \frac{1250\pi}{x} - \frac{\pi x}{2}$$

$$C = 100(2y) + 125(2\pi x)$$

$$C = 200\left(\frac{1250\pi}{x} - \frac{\pi x}{2}\right) + 250\pi x$$

$$C = \frac{250000\pi}{x} - 100\pi x + 250\pi x$$

$$C = \frac{250000\pi}{x} + 150\pi x$$

(ii) $\frac{dC}{dx} = -\frac{250000\pi}{x^2} + 150\pi = 0$

$$150\pi = \frac{250000\pi}{x^2}$$

$$x^2 = \frac{5000}{3}$$

$$x = 40.8, \quad -40.8 \text{ (NA since } x > 0 \text{)}$$

Using 1st Derivative Test,

x	40.8^-	40.8	40.8^+
$\frac{dC}{dx}$	-ve	0	+ve
Outline			

$\therefore x = 40.8$ gives a minimum value of C .

Alternatively, using 2nd Derivative test,

$$\frac{d^2C}{dx^2} = \frac{500000\pi}{x^3} = 23.1 > 0$$

$\therefore x = 40.8$ gives a minimum value of C .

4(i) $9x - 3 = \frac{4}{x} - x$

$$9x^2 - 3x = 4 - x^2$$

$$10x^2 - 3x - 4 = 0$$

$$x = \frac{3 \pm \sqrt{3^2 - 4(10)(-4)}}{2(10)}$$

$$x = \frac{3 \pm 13}{20} = \frac{4}{5}, -\frac{1}{2} \quad (\text{NA since } x > 0)$$

(ii) $\int \left(\frac{4}{x} - x \right) dx = 4 \ln x - \frac{1}{2} x^2 + C$

(iii) Line $y = 9x - 3$ cuts the x -axis at $\left(\frac{1}{3}, 0 \right)$ and curve $y = \frac{4}{x} - x$ cuts the x -axis at $(2, 0)$

Area of shaded region

$$= \int_{\frac{1}{3}}^{\frac{4}{5}} (9x - 3) dx + \int_{\frac{4}{5}}^2 \left(\frac{4}{x} - x \right) dx$$

$$= \left[\frac{9}{2} x^2 - 3x \right]_{\frac{1}{3}}^{\frac{4}{5}} + \left[4 \ln x - \frac{1}{2} x^2 \right]_{\frac{4}{5}}^2$$

$$= 2.97$$

$$5(i) \quad \frac{dy}{dx} = \frac{2}{2x-1} - 2x = 0$$

$$\frac{1}{2x-1} = x$$

$$2x^2 - x - 1 = 0$$

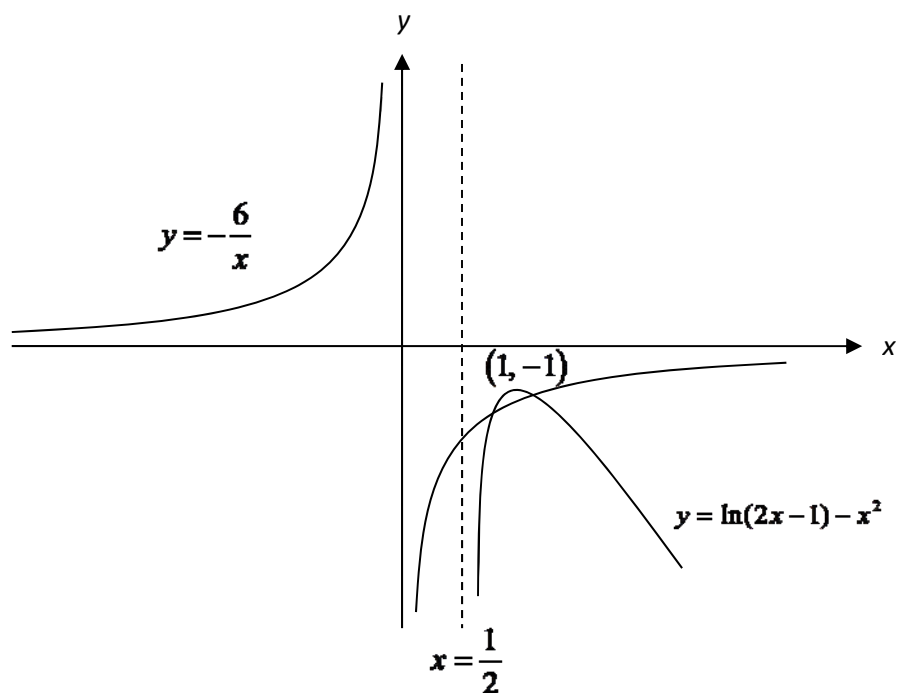
$$(2x+1)(x-1) = 0$$

$$x = 1 \quad \text{or} \quad x = -\frac{1}{2} \quad (\text{NA since } x > \frac{1}{2})$$

$$\text{When } x = 1, \quad y = \ln(2-1) - 1 = -1$$

$\therefore$ The stationary point of C is $(1, -1)$.

(ii)



$$(iii) \quad x(\ln(2x-1) - x^2) + 6 = 0$$

$$\ln(2x-1) - x^2 = -\frac{6}{x}$$

Sketch the graph of $y = -\frac{6}{x}$.

From the graph, there are 2 points of intersection. Hence, the equation has 2 real roots.

- 6(i)** A list of all the pupils in the school is first obtained and each pupil is assigned a number from 1 to 1500.

A starting point is randomly chosen between 1 - 10, say Pupil 2.

$$k = \frac{1500}{150} = 10$$

For example, if Pupil 2 is chosen, then every subsequent 10th pupil, i.e. Pupil 12, 22, 32, ... is chosen until the sample of 150 pupils is obtained.

- (ii)** Select the number of students from each strata based on the numbers in the table below

	Male	Female
Year 1	35	45
Year 2	30	40

Select the number of students in each strata by using simple random sampling.

- (iii)** Stratified sampling is more appropriate because the sample collected is proportionately representative of the population.

- 7 (i) Since R and S are independent,

$$P(R|S) = P(R) = 0.4$$

- (ii) Since S and C are independent,

$$P(S \cap C) = P(S)P(C) = 0.135$$

$$P(S \cup C) = P(S) + P(C) - P(S \cap C) = 0.615$$

- (iii) Since R and C are independent,

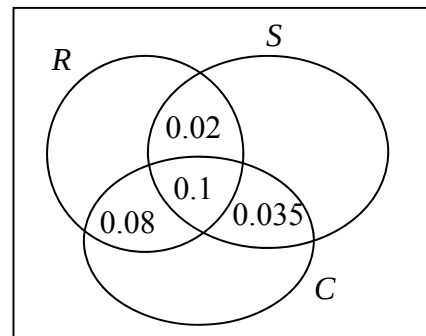
$$P(R \cap C) = P(R)P(C) = 0.18$$

$$P(S \cup C \cup R)$$

$$\begin{aligned} &= P(S \cup C) + P[(S \cup C)' \cap R] \\ &= 0.615 + (0.4 - 0.08 - 0.1 - 0.02) \\ &= 0.815 \end{aligned}$$

$$P(R' \cap S' \cap C')$$

$$= 1 - P(S \cup C \cup R) = 0.185$$



$$\text{Required probability} = (0.02 + 0.08 + 0.035)^2 = 0.018225 \approx 0.0182 \text{ (3sf)}$$

Alternatively

$X \sim$ number of adults who exercise exactly two out the three events, out of 2

$$X \sim B(2, 0.135)$$

$$P(X = 2) = 0.0182(3sf)$$

8. Let $X \sim$ number of deals clinched, out of 20

$$X \sim B(20, 0.2)$$

(i) $P(X > 3) = 1 - P(X \leq 3) = 0.58855 \approx 0.589$ (3sf)

(ii) $P(\mu - \sigma < X < \mu + \sigma)$
 $= P(2.2111 < X < 5.7888)$
 $= P(3 \leq X \leq 5)$
 $= P(X \leq 5) - P(X \leq 2)$
 $= 0.59812 \approx 0.598$ (3sf)

(iii) $E(X) = 20 \times 0.2 = 4$

$$\text{Var}(X) = 20 \times 0.2 \times (1 - 0.2) = 3.2$$

As $n = 52$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(4, \frac{4}{65}\right)$ approx.

$$P(\bar{X} \leq 4.5) = 0.97808 \approx 0.978$$

Let $Y \sim$ number of deals clinched, out of 80

$$Y \sim B(80, 0.2)$$

Since $n = 80$ is large, $np = 16 > 5$, $n(1 - p) = 64 > 5$

$Y \sim N(16, 12.8)$ approx.

$$P(Y > 15) = P(Y > 15.5) \quad (\text{by continuity correction})$$
$$= 0.55557 \approx 0.556 \text{ (3sf)}$$

9. $X \sim$ mass of an unfrosted cupcake. $X \sim N(87, 9)$

$Y \sim$ mass of frosting. $Y \sim N(27, 4)$

(i) $P(X > 86) = 0.63055 \approx 0.631$ (3sf)

(ii) $W = X + Y$

$$W \sim N(114, 13)$$

$$P(110 < W < 120) = 0.81833 \approx 0.818$$

(iii) The mass of an unfrosted cupcake and the mass of frosting are assumed to be independent.

(iv) $P(3W > 340) = P(W > \frac{340}{3}) = 0.57335 \approx 0.573$

Alternatively,

$$3W \sim N(342, 117)$$

$$P(3W > 340) = 0.57335 \approx 0.573$$

$$\bar{W} \sim N\left(114, \frac{13}{n}\right)$$

$$P(\bar{W} < 112) \leq 0.001$$

n	$P(\bar{W} < 112)$
31	0.00101
32	0.00085
33	0.00071

Least $n = 32$

Alternatively

$$P(\bar{W} < 112) \leq 0.001$$

$$P\left(Z < \frac{112 - 114}{\sqrt{\frac{13}{n}}}\right) \leq 0.001$$

$$\frac{-2}{\sqrt{\frac{13}{n}}} \leq -3.0902$$

$$2\sqrt{n} \geq 11.142$$

$$n \geq 31.036$$

Least $n = 32$

10 (i) $\bar{x} = \frac{\sum(x-25)}{80} + 25 = 25.3$

$$s^2 = \frac{1}{80-1} \left[230 - \frac{(24)^2}{80} \right] = 2.8203 \approx 2.82 \text{ (3sf)}$$

(ii) An estimate of a population parameter is an unbiased estimate if its expected value is equal to the true value of the population parameter.

(iii) Test $H_0 : \mu = 25$ vs $H_1 : \mu > 25$

Under H_0 , since $n = 80$ is large, by Central Limit Theorem,

$$\bar{X} \sim N\left(25, \frac{2.8203}{80}\right) \text{ approx.}$$

Level of significance: 5%.

Critical region: $z \geq 1.6449$

$$\text{Standardized test statistic: } z = \frac{\bar{x} - \text{"claimed value"}}{s/\sqrt{n}} = \frac{25.3 - 25}{\sqrt{\frac{2.8203}{80}}} = 1.5978$$

From GC, $p\text{-value} = 0.055045 > 0.05$

Since the $p\text{-value}$ is more than the level of significance, we do not reject H_0 .

There is sufficient evidence at 5% level of significance to doubt the owner's claim.

(iv) Test $H_0 : \mu = 25$ vs $H_1 : \mu > 25$

$$\text{Standardised Test statistic, } z = \frac{\bar{x} - \mu}{\sqrt{\frac{\sigma^2}{n}}} = \frac{k - 25}{\sqrt{\frac{2}{100}}}$$

Critical region: $z \geq 1.2816$

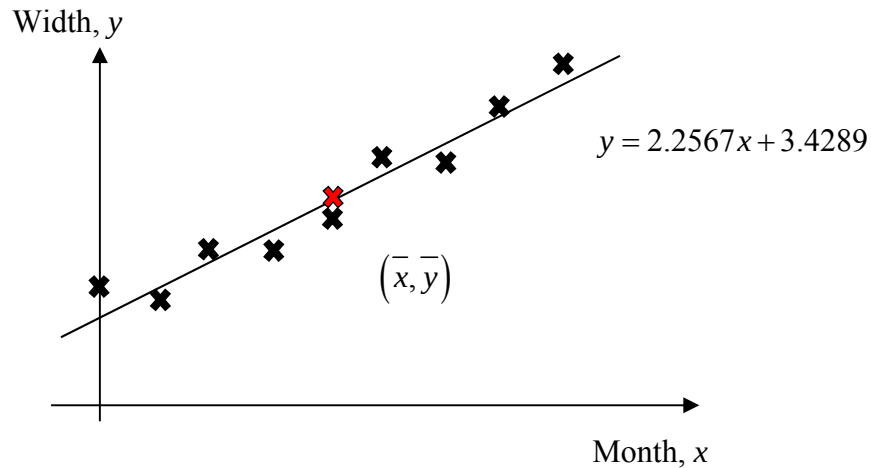
For fish farm's claim to be valid, H_0 rejected.

$$\frac{k - 25}{\sqrt{\frac{2}{100}}} \geq 1.2816$$

$$k \geq 25.18 \text{ (2dp)}$$

- 11** To assess whether a linear relationship between the two variables exist.

(i)



(ii) $y = 2.2567x + 3.4289 \approx 2.26x + 3.43$ (3sf)

(iii) Using GC $(\bar{x}, \bar{y}) = (4, 12.5)$ (use 2 variable stats)

(iv) $r = 0.983$ (3sf) is close to 1, there is a strong positive linear correlation between the width and the age (in months)

(v) $y = 2.2567(4.5) + 3.4289 = 13.584 \approx 13.6$

The width of a 4.5 month old stingray will be 13.6 cm. Since $x = 4.5$ is within the data set, it is an interpolation, and since $r = 0.983$ is close to 1, there is a strong positive linear correlation between the width and the age (in months), the estimate is reliable.